

FFmpeg Converter GUI

Prerequisites & Installation Guide · Version 2.6

Before you can use FFmpeg Converter GUI, two things need to be in place: the **FFmpeg** binary on your system, and (strongly recommended) a supported **NVIDIA GPU** for hardware-accelerated encoding.

1. FFmpeg 8.0 or newer (required)

FFmpeg Converter GUI is a graphical front-end for FFmpeg. The actual video conversion is done by the FFmpeg command-line tool, which must be installed separately.

Why version 8.0 or newer? The GUI uses modern FFmpeg syntax such as `-init_hw_device cuda:N` for multi-GPU selection and `-fps_mode cfr` for Mediathek-Safe mode. These options are not available in older FFmpeg versions and the GUI will refuse to start the encoding process if FFmpeg is too old.

Installation:

- Download a recent build from <https://ffmpeg.org/download.html> (Windows users: the “gyan.dev” or “BtbN” builds are recommended).
- Extract the archive and either add the **bin/** folder to your system **PATH**, or copy **ffmpeg.exe** and **ffprobe.exe** into the same folder as the GUI executable.
- Verify the installation by opening a terminal and running `ffmpeg -version` — the first line must report version 8.0 or higher.

2. NVIDIA GPU with NVENC (strongly recommended)

The GUI is designed around **NVIDIA NVENC**, the dedicated hardware encoder built into modern GeForce and RTX cards. NVENC dramatically speeds up the conversion process compared to software (CPU-only) encoding — typically by a factor of **5x to 20x**, depending on the codec, resolution, and the GPU generation.

Without a supported GPU the GUI still works, but encoding will fall back to the CPU and even short clips can take many minutes per minute of video. For 4K material, hardware encoding is essentially mandatory.

Recommended hardware:

- For **AV1** encoding: GeForce RTX 40-series or newer (Ada Lovelace; first NVENC generation with AV1 support).
- For **H.265 (HEVC)** and **H.264**: Any GeForce GTX 10-series or newer with NVENC works; newer generations are faster and produce better quality at the same bitrate.
- A current NVIDIA driver is required (the GUI uses `nvidia-smi` to detect available GPUs).

3. Quick checklist

Step	How to verify
FFmpeg 8.0+ in PATH	Run <code>ffmpeg -version</code> in a terminal
ffprobe 8.0+ available	Run <code>ffprobe -version</code> in a terminal
NVIDIA driver installed	Run <code>nvidia-smi</code> in a terminal
GPU recognised by GUI	Open the app: the "Encoder" dropdown lists your GPU(s)